



Incorporating Auxiliary Variables to Improve the Efficiency of Time-Varying Treatment Effect Estimation

Jieru Shi, Zhenke Wu & Walter Dempsey

To cite this article: Jieru Shi, Zhenke Wu & Walter Dempsey (2026) Incorporating Auxiliary Variables to Improve the Efficiency of Time-Varying Treatment Effect Estimation, Journal of the American Statistical Association, 121:553, 300-311, DOI: [10.1080/01621459.2025.2516197](https://doi.org/10.1080/01621459.2025.2516197)

To link to this article: <https://doi.org/10.1080/01621459.2025.2516197>



© 2025 The Author(s). Published with license by Taylor & Francis Group, LLC.



[View supplementary material](#)



Published online: 05 Aug 2025.



[Submit your article to this journal](#)



Article views: 1276



[View related articles](#)



[View Crossmark data](#)



Citing articles: 2 [View citing articles](#)

Incorporating Auxiliary Variables to Improve the Efficiency of Time-Varying Treatment Effect Estimation

Jieru Shi, Zhenke Wu, and Walter Dempsey

Department of Biostatistics, University of Michigan, Ann Arbor, MI

ABSTRACT

Contextual sensing and delivery of digital interventions to improve health outcomes have gained significant traction in behavioral and psychiatric studies. Micro-randomized trials (MRTs) are a common experimental design for obtaining data-driven evidence on the effectiveness of digital interventions where each individual is repeatedly randomized to receive treatments over numerous time points. Throughout the study, individual characteristics and contextual factors around randomization are collected, with some prespecified as moderators for assessing time-varying causal effect moderation. However, many additional measurements beyond these moderators often go underutilized. Some of these may influence treatment randomization or known to strongly moderate the treatment effect. Incorporating such auxiliary information into the estimation procedure can reduce chance imbalances and improve asymptotic estimation efficiency. In this work, we propose a method to adjust for auxiliary variables in consistently estimating time-varying intervention effects. The approach can also be extended to include post-treatment auxiliary variables when evaluating lagged treatment effects. Under specific conditions, local efficiency gains are guaranteed. We demonstrate the method's utility through simulation studies and an analysis of data from the Intern Health Study (NeCamp et al.). Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

ARTICLE HISTORY

Received August 2023
Accepted May 2025

KEYWORDS

Asymptotic efficiency; Causal inference; Covariate adjustment; Micro-randomized trials; Mobile health; Moderation effect

1. Introduction

The use and development of mobile interventions is experiencing rapid growth. In “just-in-time” mobile interventions (Nahum-Shani et al. 2018), treatments are provided via mobile devices that are intended to help individuals make healthy decisions “in the moment” and thus have a proximal, near-term impact on health outcomes. Micro-randomized trials (MRTs; Klasnja et al. 2015; Dempsey et al. 2015) provide data to guide the development of such mobile interventions (Free et al. 2013), where each participant is sequentially randomized to different treatment options at possibly hundreds to thousands of occasions. In an MRT, measurements of individual characteristics, context, and response to treatments are also collected repeatedly, often through passive sensing or self-report. Some of these measurements may be identified in advance as primary moderators of interest when assessing time-varying treatment effect, while the others are referred to as *auxiliary variables*. MRTs admit the definition of a class of causal estimands known as “causal excursion effects” (Boruvka et al. 2018), and the precise estimation of these effects help inform domain scientists whether to include the treatment in an mHealth intervention package.

The weighted and centered least squares (WCLS) method (Boruvka et al. 2018) is the state-of-the-art for estimating causal excursion effects. It can be framed as a two-stage regression approach. In the first stage, the observed history is trained to pre-

dict a proximal or possibly lagged outcome. In the second stage, the debiased outcomes are regressed onto a low-dimensional treatment effect model for interpretable results. While this procedure ensures consistent and asymptotically normal estimates of the causal excursion effect, it does not take full advantage of the rich auxiliary information available over time, which can lead to suboptimal precision. In many cases, certain pre-treatment auxiliary variables may be known to strongly moderate the treatment effect, or when studying lagged causal effects, post-treatment auxiliary variables can often be highly predictive. Incorporating these variables into estimation can potentially reduce noise and improve the precision of causal estimates, allowing scientists to make better use of the available data.

Similar issues have been thoroughly explored in the context of single-point treatments in randomized controlled trials (RCTs), particularly when estimating the average treatment effect (ATE). Covariate adjustment is a key technique in this setting, addressing chance imbalances in baseline covariates and maintaining correct Type I error rates when covariates are part of the randomization process (Kahan et al. 2014; Tackney et al. 2023). Moreover, incorporating “mean-centered” covariates into regression analyses has been shown to improve asymptotic precision when all treatment-covariate interactions are included in the model (Lin 2013; Negi and Wooldridge 2021; Ye et al. 2023). In MRTs, auxiliary variables may be predictive of the repeated outcomes and be used to improve the efficiency of estimating

CONTACT Jieru Shi  herashi@umich.edu  Department of Biostatistics, University of Michigan, 1415 Washington Heights, Ann Arbor, MI 48109..

 Supplementary materials for this article are available online. Please go to www.tandfonline.com/r/JASA.

© 2025 The Author(s). Published with license by Taylor & Francis Group, LLC.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

the causal excursion effects. However, incorporating these variables presents challenges due to their time-varying nature in the context of sequential randomization studies. A critical question this article seeks to address is “How can ‘mean-centered’ time-varying auxiliary variables be incorporated to avoid causal bias and guarantee either efficiency gains or no efficiency loss?”

We will show in this article that “centering” of a time-varying auxiliary variable is not uniquely defined in our more general context. To see this, consider time-specific mean centering, which naturally extends Lin’s (2013) covariate adjustment to the time-varying setting. In an MRT with many decision points, this choice will lead to a large number of nuisance centering parameters. Moreover, this choice only guarantees consistency for fully marginal causal effects. We will show that for many moderation analyses, this choice will lead to causal bias. A low-dimensional alternative such as a global weighted mean might be more appealing; however, an incorrect choice will again lead to biased estimates. This motivates our proposal of a practical and theoretically sound framework for incorporating auxiliary variables into the estimating procedure with the goal of improving efficiency while ensuring consistent estimation.

Main Contributions. In this article, we propose the Auxiliary-variable Adjusted WCLS (A2-WCLS) method to estimate causal excursion effects in MRTs. Our approach consists of two key components: (a) Pre-treatment auxiliary variable adjustment, where we establish an orthogonality condition that ensures consistent causal parameter estimation while maintaining a low-dimensional estimating equation for nuisance centering parameters, preventing their prohibitive growth with increasing time points. (b) Post-treatment auxiliary variable adjustment, where we show how to incorporate highly predictive post-treatment variables without introducing bias, further enhancing estimation precision. We prove that A2-WCLS guarantees a local efficiency gain when the causal effect model is correctly specified. Extensive simulations demonstrate the significant efficiency gains achieved with this approach. Finally, we apply A2-WCLS to a real-world mobile health study, demonstrating how the improved estimation efficiency allows domain scientist to have more precise evaluation of the magnitude and duration of time-varying treatment effects.

The article is organized as follows: Section 2 reviews current methods for MRTs and covariate adjustment in single-time-point RCTs, highlighting challenges in time-varying settings. Section 3 introduces our strategy for incorporating auxiliary variables and outlines its asymptotic properties. Section 4 presents the main theorem on efficiency improvements from proposed auxiliary adjustment. Section 5 summarizes the statistical properties of the proposed method. Section 6 includes simulations comparing different estimators and showcasing efficiency gains, while Section 7 demonstrates these gains using data from the Intern Health Study (NeCamp et al. 2020). The article concludes in Section 8, with technical proofs and additional results provided in the supplementary materials.

2. Preliminaries

For a given individual j , let $A_{t,j}$ denote the treatment at the t th occasion, where $t = 1, \dots, T; j = 1, \dots, N$. We define $Y_{t+\Delta,j}$ as

the outcome measured at future decision points within a fixed window $\Delta \geq 1$. In the special case where $\Delta = 1$, $Y_{t+1,j}$ represents the subsequent proximal response. Individual j ’s contextual information at the t th treatment occasion is represented by $X_{t,j}$, which may contain summaries of historical context, treatments, or response measurements. For example, before each treatment occasion, the individual might report their current mood status, then the vector $X_{t,j}$ could contain this measurement or, with previous measurements, variation or change in mood. Hereafter, we may omit the subscript j for ease of presentation unless otherwise clarified. The overbar “ $\bar{\cdot}$ ” is used to denote a sequence of random variables or realized values up to a specific treatment occasion. For example, $\bar{A}_{t-1} = (A_1, \dots, A_{t-1})$ denotes the sequence of treatments up to and including decision time $t - 1$. Information accrued up to treatment occasion t is represented by the history $H_t = (\bar{A}_{t-1}, \bar{X}_t)$. For simplicity, we present the main results in this article under a binary treatment $A_t \in \{0, 1\}$ which follows a known sequence of randomization probabilities p_t that may depend on the complete observed history, denoted $\{p_t = p_t(A_t|H_t)\}_{t=1}^T$. In the following, random variables or vectors are denoted with uppercase letters; lowercase letters denote their realized values. In addition, we assume that the longitudinal data are independent and identically distributed across N individuals. Note that this assumption would be violated, if, for example, some of the treatments are used to enhance social support between individuals in the study (Liao et al. 2016).

2.1. Causal Excursion Effects and Existing Inferential Methods

Many treatments are designed to influence an individual in the short-term (or “proximal”) (Heron and Smyth 2010). To answer questions related to the causal effect of time-varying treatments on the proximal or lagged response, we focus on a class of estimands referred to as “causal excursion effects”. In particular, a causal excursion effect is a function of the decision point t and a set of moderators S_t , marginalizing over all other observed and unobserved variables (Boruvka et al. 2018; Qian et al. 2021). We provide formal definitions below using potential outcomes (Rubin 1978; Robins 1986).

Let $Y_{t+\Delta}(\bar{a}_{t+\Delta-1})$ denote the potential outcome for the proximal response under treatment sequence $\bar{a}_{t+\Delta-1}$. Let $S_t(\bar{a}_{t-1})$ denote the potential outcome for a time-varying effect moderator which is a deterministic function of the potential history up to time t , $H_t(\bar{a}_{t-1})$. The Δ -lag causal excursion effect is defined with respect to a reference distribution $\{\pi_{t+u}(a_{t+u}|H_{t+u})\}_{u=1}^{\Delta-1}$, that is, the treatment randomization probabilities from $t+1$ to $t+\Delta-1$. This generalization contains previous definitions such as the natural lagged effects (Boruvka et al. 2018) where $\pi_{t+u} = p_{t+u}$ and deterministic choices such as $a_{t+1:(t+\Delta-1)} = \mathbf{0}$ (Dempsey et al. 2020; Qian et al. 2021), where $\pi_{t+u} = \mathbf{1}\{a_{t+u} = 0\}$ and $\mathbf{1}\{\cdot\}$ is the indicator function. Then the Δ -lag causal excursion effect is defined as follows:

$$\begin{aligned} \beta_{\mathbf{p},\pi}(t + \Delta - 1; s) = & \mathbb{E}_{\mathbf{p},\pi} [Y_{t+\Delta}(\bar{A}_{t-1}, A_t = 1) \\ & - Y_{t+\Delta}(\bar{A}_{t-1}, A_t = 0) \mid S_t(\bar{A}_{t-1}) = s]. \end{aligned} \quad (1)$$

Equation (1) is defined with respect to both $\mathbf{p}$ and π , the distributions of treatment sequence $\bar{A}_{t-1}$ and $A_{t+1:(t+\Delta-1)}$, respectively. We follow common approaches in observational mobile health studies, where analyses are performed marginally over $\mathbf{p}$ and π , such as GEEs (Liang and Zeger 1986). To express the causal excursion effect $\beta_{\mathbf{p},\pi}(t;s)$ in terms of the observed data, we assume positivity, consistency, and sequential ignorability (Robins 1994, 1997):

- Assumption 2.1 (Identification).** • Consistency: $\{Y_{t+\Delta,j}(\bar{A}_{t+\Delta-1,j}), X_{t,j}(\bar{A}_{t-1,j}), A_{t,j}(\bar{A}_{t-1,j})\} = \{Y_{t+\Delta,j}, X_{t,j}, A_{t,j}\}$, for each $t \leq T$ and j , that is, observed values equal the corresponding potential outcomes;
- Positivity: if the joint density $\mathbb{P}\{H_t = h_t\}$ is greater than zero, then $\mathbb{P}(A_t = a | H_t = h_t) > 0$, $a = 0, 1$;
 - Sequential ignorability: For each $t \leq T$, the potential outcomes $\{Y_{t+\Delta}(\bar{a}_t), X_{t+1}(\bar{a}_t), A_{t+1}(\bar{a}_t), \dots, Y_{T+1}(\bar{a}_T)\}$ are independent of A_t conditional on the observed history H_t .

We consider the setting in which the potential outcomes are iid over individuals according to a distribution $\mathcal{P}$. Under Assumption 2.1, the Δ -lag causal excursion effect defined in (1) can be expressed in terms of observable quantities as (Shi, Wu, and Dempsey 2023):

$$\beta_{\mathbf{p},\pi}(t + \Delta - 1; s) = \mathbb{E}[\mathbb{E}[W_{t,\Delta-1} Y_{t+\Delta} | A_t = 1, H_t] - \mathbb{E}[W_{t,\Delta-1} Y_{t+\Delta} | A_t = 0, H_t] | S_t = s], \quad (2)$$

where $W_{t,u} = \prod_{s=1}^u \pi_{t+s}(A_{t+s}|H_{t+s})/p_{t+s}(A_{t+s}|H_{t+s})$. Assuming $\beta_{\mathbf{p},\pi}(t + \Delta; s) = f_t(s)^\top \beta_0^*$, where $f_t(s) \in \mathbb{R}^q$ is a q -dimensional feature vector with a summary of observed information depending only on the effect moderator value s and decision point t , the WCLS criterion for Δ -lag causal effect is to minimize the following expression (Boruvka et al. 2018):

$$\mathbb{P}_N \left[\sum_{t=1}^{T-\Delta+1} W_{t,\Delta-1} W_t \times (Y_{t+\Delta} - g_t(H_t)^\top \alpha_0 - (A_t - \tilde{p}_t(1 | S_t)) f_t(S_t)^\top \beta_0)^2 \right]. \quad (3)$$

where $\mathbb{P}_N$ is an operator denoting the sample average, $W_t = \tilde{p}_t(A_t | S_t)/p_t(A_t | H_t)$ is a weight where the numerator is an arbitrary function with range (0, 1) that only depends on S_t , and $g_t(H_t) \in \mathbb{R}^d$ are d control variables chosen to help reduce noise and to construct more powerful test statistics. See Boruvka et al. (2018) for more details on the estimand definition and consistency, asymptotic normality, and robustness properties of the WCLS estimation method.

In addition to studying continuous outcomes, there are also many MRTs concerning longitudinal binary outcomes. Qian et al. (2021) proposed an estimator of the marginal excursion effect (EMEE) by defining a log relative risk model for the causal excursion effect, and we present a detailed review in Appendix L.

2.2. Literature Review: Covariate Adjustment

The RCT methodology literature has extensively examined the use of covariate adjustment to improve the precision of infer-

ence on ATEs. Consider, for example, an RCT with a binary randomized treatment denoted A , and N subjects in the study are the population of interest. Neyman (1923) showed that the difference-in-means estimator is unbiased for the ATE. Fisher (1936) proposed to use the ordinary least squares (OLS) adjusted estimator of ATE, which is the estimated coefficient on treatment A in the OLS regression of Y on $\{1, A, X_0\}$. This approach aims to leverage prognostic information in baseline covariates to improve estimation efficiency. However, Freedman (2008) demonstrated that in specific scenarios, such adjustment could potentially harm asymptotic precision.

Lin (2013) argued that in sufficiently large samples, the issues related to “adjustment worsening asymptotic precision” Freedman (2008) raised are either minor or easily fixed. In addition, Lin (2013) shows that OLS adjustment with a full set of treatment and covariate interactions improves or does not hurt asymptotic precision, even when the regression model is incorrect. The robust Eicker-Huber-White (Eicker 1967; Huber 1967; White 2014) variance estimator (Freedman 2006) is consistent or asymptotically conservative (regardless of whether the interactions are included) in estimating the true asymptotic variance. More results on the asymptotic precision of treatment effect can be found in papers by Yang and Tsiatis (2001), Tsiatis et al. (2008), Vansteelandt and Didelez (2018), and Negi and Wooldridge (2021). More complex randomization schemes have led to recent developments in covariate adjustment methods. For example, Su and Ding (2021) extended the theory to cluster-randomized experiments. Ye et al. (2023) used a model-assisted approach for covariate adjustment and generalized to linear contrast, risk ratios, and odds ratios. Their conclusions are based on an asymptotic theory that provides a clear picture of how covariate-adaptive randomization and regression adjustment alter statistical efficiency.

Guo and Basse (2023) and Ye et al. (2023) both proposed checklists of ideal properties for covariate adjustment methods. As we consider time-varying treatment effect estimation, we build upon their lists of the ideal properties that the auxiliary variable adjustment method should have: (a) valid statistical inference; (b) robust estimation; (c) strong finite-sample performance; (d) wide applicability; and (e) computational simplicity. In the following sections, we propose and evaluate a general method that satisfies the five ideal properties for performing auxiliary variable adjusted WCLS, termed “A2-WCLS”, for improving efficiency of estimating causal excursion effects in longitudinal data with time-varying treatments.

3. Estimand and Inferential Method

We focus on two key causal estimands in longitudinal settings: the time-specific marginal causal effect and the time-smoothed moderated causal effect. We develop a general estimation framework that incorporates auxiliary variables to improve the estimation of causal excursion effects. The main theoretical findings on asymptotic efficiency comparisons are summarized in Table 1. The results here help answer the earlier question of whether using auxiliary variables is beneficial and how to effectively include them in the estimation process. To begin, we make a parametric assumption about the causal parameter of interest:

Table 1. Summary of asymptotic efficiency comparisons under [Assumptions 2.1](#) and [3.1](#) among the unadjusted estimator, WCLS (Boruvka et al. 2018), the covariate adjustment method by Lin (2013), and the proposed A2-WCLS estimators for both time-specific and smoothed estimands.

Estimator	Assumptions	Asymptotic efficiency comparison	
<i>Time-specific marginal causal effect (Section 3.1)</i>			
Unadjusted	–	Baseline	$p_t = 0.5$: improved or equal; otherwise: not guaranteed Improved or equal
WCLS	–	Lemma 3.2	
Lin (2013)	–	Lemma 3.3	
<i>Time-smoothed moderated causal effect (Section 3.2)</i>			
WCLS	–	Baseline	Improved or equal Equal
A2-WCLS	Condition 3.4, Assumptions 4.1, 4.2	Theorem 4.3	
	Condition 3.4, Proposition 4.4	Proposition 4.4	

Assumption 3.1. Assume the causal excursion effect $\beta_{p,\pi}(t + \Delta - 1; s) = f_t(s)^\top \beta_0^*$, where $f_t(s) \in \mathbb{R}^q$ is a feature vector comprised of a q -dimensional summary of observed information depending only on state s and decision point t .

This parametric assumption assumes a correct specification of the causal effect; however, when model misspecification occurs, we can still interpret the proposed linear form as an L_2 projection of the true causal excursion effect onto the space spanned by a q -dimensional feature vector $f_t(s)$ that only depends on t and s (Shi, Wu, and Dempsey 2023). The choice between these interpretations, whether it is a correctly specified causal effect or a projection, reflects a bias-variance tradeoff. In practical applications, the projection interpretation ensures a well-defined parameter with practical interest (Dempsey et al. 2020).

3.1. Time-Specific Causal Excursion Effect Estimation

To establish a clear connection with previous research on covariate adjustment, we start with the *time-specific causal excursion effect estimand* with $\Delta = 1$, which shares similarities with the ATE estimand in RCTs. Recall that the ATE estimand is typically defined as the linear difference between the expected outcomes under different treatment allocations, that is, $\mathbb{E}[\mathbb{E}[Y|X, A = 1] - \mathbb{E}[Y|X, A = 0]]$. In the context of MRTs, the time-specific marginal causal effect estimand is defined as $\beta_p(t) = \mathbb{E}[\mathbb{E}_p[Y_{t+1}|H_t, A_t = 1] - \mathbb{E}_p[Y_{t+1}|H_t, A_t = 0]]$, with the additional time stamp and controlling for the treatment randomization probability, which depends on the observed history. Moving forward, we drop the subscript p to simplify the notation. The time-specific causal excursion effect is estimated by setting $\beta(t; S_t) = \beta_{0,t}^*$ in [Assumption 3.1](#), with $S_t = \emptyset$, allowing for a nonparametric estimation over time. The treatment randomization probability is denoted as $p_t(A_t|H_t) = p_t$. In this subsection, the phrase “more efficient” explicitly refers to an estimator of $\beta_{0,t}^*$ that has a smaller asymptotic variance.

3.1.1. The Unadjusted and the WCLS Estimator

Analysis using solely the data from time-varying outcomes and treatments, $\{Y_{t+1,j}, A_{t,j}\}_{j=1}^T$, $j = 1, \dots, N$, without adjusting for auxiliary variables yields the *unadjusted estimator*. The unadjusted estimator $\hat{\beta}_{0,t}^U$ can be obtained by minimizing the following objective function:

$$\mathbb{P}_N \left[\sum_{t=1}^T (Y_{t+1} - \alpha_{0,t} - (A_t - p_t) \beta_{0,t})^2 \right]. \quad (4)$$

To potentially improve the precision of causal parameter estimates using auxiliary variables, we could use the objective function presented in (3) with $\beta(t; S_t) = \beta_{0,t}^*$ and mean-centered control variables $\tilde{g}_t(H_t) = g_t(H_t) - \mathbb{E}[g_t(H_t)]$. The WCLS estimator, $\hat{\beta}_{0,t}^{WCLS}$, is thus obtained by minimizing the following objective function:

$$\mathbb{P}_N \left[\sum_{t=1}^T (Y_{t+1} - \alpha_{0,t} - \tilde{g}_t(H_t)^\top \alpha_{1,t} - (A_t - p_t) \beta_{0,t})^2 \right]. \quad (5)$$

Both estimators, $\hat{\beta}_{0,t}^U$ and $\hat{\beta}_{0,t}^{WCLS}$, are consistent and asymptotically normal (Boruvka et al. 2018). This raises a natural question: does including the additional term consistently improve precision? If not, under what conditions might the asymptotic variance of the causal parameter estimates increase when using $\hat{\beta}_{0,t}^{WCLS}$ instead of $\hat{\beta}_{0,t}^U$? To address this, we present the following lemma:

Lemma 3.2 (Asymptotic Variance Comparison between $\hat{\beta}_{0,t}^{WCLS}$ and $\hat{\beta}_{0,t}^U$). Denote $\beta_{1,t}^* = \Sigma(g_t(H_t))^{-1} \mathbb{E}[Y_{t+1}(A_t - p_t)\tilde{g}_t(H_t)] \in \mathbb{R}^d$, where $\Sigma(g_t(H_t)) = \mathbb{E}[\tilde{g}_t(H_t)\tilde{g}_t(H_t)^\top] \in \mathbb{R}^{d \times d}$. The difference in the asymptotic variance between the WCLS estimator $\hat{\beta}_{0,t}^{WCLS}$ and the unadjusted estimator $\hat{\beta}_{0,t}^U$ is $-p_t(1 - p_t)\alpha_{1,t}^* \Sigma(g_t(H_t))(\alpha_{1,t}^* + 2(1 - 2p_t)\beta_{1,t}^*)$.

Here, $\alpha_{1,t}^*$ denotes the true coefficient of the variables $\tilde{g}_t(H_t)$ in (5), while $\beta_{1,t}^*$ represents their true moderation effect size. The least-squares estimator for $\beta_{1,t}^*$ is obtained by minimizing the objective function in (6). If $p_t = 1/2$, the adjustment is neutral or beneficial. However, if $g_t(H_t)$ is positively correlated with the outcome and strongly moderates the treatment effect (i.e., $\alpha_{1,t}^* > 0$ and $|\beta_{1,t}^*|$ is large), an adjustment may increase the asymptotic variance when $\beta_{1,t}^* > 0$ and $p_t > 1/2$. Thus, the WCLS criterion in (5) does not always guarantee efficiency gains over the unadjusted estimator in (4), consistent with Freedman’s (2008) findings on covariate adjustment in RCTs.

3.1.2. Proposed and More Efficient Estimator

To ensure improved estimation efficiency, we propose an alternative adjustment method for time-varying auxiliary

variables. This approach integrates Lin's (2013) idea into the WCLS framework, yielding an estimator for the causal excursion effect obtained by minimizing:

$$\mathbb{P}_N \left[\sum_{t=1}^T (Y_{t+1} - \alpha_{0,t} - \tilde{g}_t(H_t)^\top \alpha_{1,t} - (A_t - p_t) (\beta_{0,t} + \tilde{g}_t(H_t)^\top \beta_{1,t}))^2 \right]. \quad (6)$$

Equation (6) is a strict extension of the proposal in Lin (2013); we denote the minimizer as $(\hat{\alpha}_{0,t}, \hat{\alpha}_{1,t}, \hat{\beta}_{0,t}^L, \hat{\beta}_{1,t}^L)$. Lemma 3.3 below establishes the asymptotic efficiency gain of $\hat{\beta}_{0,t}^L$ over both the unadjusted $\hat{\beta}_{0,t}^U$ and the WCLS $\hat{\beta}_{0,t}^{\text{WCLS}}$ estimators.

Lemma 3.3. Suppose Assumptions 2.1 and 3.1 hold, and the randomization probability p_t is known. Given invertibility and regularity conditions, let $(\hat{\alpha}_{0,t}, \hat{\alpha}_{1,t}, \hat{\beta}_{0,t}^L, \hat{\beta}_{1,t}^L)$ minimize objective function (6):

1. $\hat{\beta}_{0,t}^L$ is consistent and asymptotically normal such that $\sqrt{N}(\hat{\beta}_{0,t}^L - \beta_{0,t}^*) \rightarrow \mathcal{N}(0, Q_t^{-1} \Sigma_t^L Q_t^{-1})$, where Q_t, Σ_t^L are defined in Appendix B;
2. $\hat{\beta}_{0,t}^L$ is at least as efficient as $\hat{\beta}_{0,t}^U$ described in (4), and the asymptotic efficiency gain is

$$\frac{(\alpha_{1,t}^* + (1 - 2p_t)\beta_{1,t}^*)^\top \Sigma(g_t(H_t)) (\alpha_{1,t}^* + (1 - 2p_t)\beta_{1,t}^*)}{p_t(1 - p_t)} + \beta_{1,t}^{*\top} \Sigma(g_t(H_t)) \beta_{1,t}^*;$$

3. $\hat{\beta}_{0,t}^L$ is at least as efficient as $\hat{\beta}_{0,t}^{\text{WCLS}}$ described in (5), and the asymptotic efficiency gain is $\frac{1-3p_t+3p_t^2}{p_t(1-p_t)} \beta_{1,t}^{*\top} \Sigma(g_t(H_t)) \beta_{1,t}^*$.

Lemma 3.3 demonstrates that incorporating mean-centered auxiliary variables in the estimation of the time-specific marginal treatment effect using the objective function in (6) guarantees an efficiency improvement over both the unadjusted and WCLS estimators. A detailed proof of Lemma 3.3 is available in Appendix B. Notably, this efficiency comparison extends beyond marginal effect to also cover moderation effect estimators with *categorical* moderators, that is, when $\beta(t; S_t) = f_t(S_t)^\top \beta_{0,t}^*$ and $f_t(S_t)$ represents categorical variables. A detailed discussion and proof can be found in Appendix C.

3.2. Auxiliary Variable Adjusted WCLS (A2-WCLS) Estimation

While estimating (6) ensures an efficiency gain, estimating centering means for $g_t(H_t)$ at each time point separately introduces a substantial number of nuisance parameters, which can be impractical given the often large time horizons in MRTs. In addition, centering the auxiliary variable at its mean is not a one-size-fits-all remedy. For example, when we are interested in moderated treatment effects (i.e., $S_t \neq \emptyset$), centering by time-specific means can introduce bias. Finally, in longitudinal studies, lagged causal effects are sometimes of primary interest, but incorporating post-treatment variables requires careful handling to avoid bias. This calls for new methodological approaches. To tackle these challenges, we propose a general

estimation framework that incorporates both pre- and post-treatment auxiliary variables when assessing time-varying effect moderation, helping to reduce chance imbalance and improve estimation precision.

Here we consider auxiliary variable adjusted estimators of the causal excursion effect, with a *primary moderator* of interest $f_t(S_t) \in \mathbb{R}^q$ and a smoothed function of the causal excursion effect model $f_t(S_t)^\top \beta_0$ as mentioned in Assumption 3.1. Previously in Section 2.1, we reviewed the estimator $\hat{\beta}_0^{\text{WCLS}}$ obtained from the WCLS criterion outlined in (3) is consistent and asymptotically normal (Boruvka et al. 2018). Our goal is to develop a practical procedure for integrating auxiliary information into the estimation process and to evaluate its asymptotic efficiency compared to that of $\hat{\beta}_0^{\text{WCLS}}$. In the context of moderation effect analysis, “greater efficiency” specifically refers to a reduction in the asymptotic variance of the estimated moderated effect for a given value of $S_t = s$, that is, a smaller asymptotic $\text{var}(f_t(s)^\top \hat{\beta}_0)$.

3.2.1. Pre-treatment Auxiliary Variable Adjustment

Let $Z_t \in \mathbb{R}^p$ represent a set of p -dimensional auxiliary variables observed before treatment allocation A_t , and $Z_t \cap f_t(S_t) = \emptyset$. For clarity, we set $\Delta = 1$ in this subsection without loss of generality. An auxiliary variable adjusted estimation method using a centering function $\mu_t(S_t) \in \mathbb{R}^p$ is developed below. Under Assumption 3.1, the Auxiliary-variable Adjusted Weighted and Centered Least Square (A2-WCLS) criterion is to minimize the following:

$$\mathbb{P}_N \left[\sum_{t=1}^T W_t \times \left(Y_{t+1} - g_t(H_t)^\top \alpha - (A_t - \tilde{p}_t(1 | S_t)) (f_t(S_t)^\top \beta_0 + (Z_t - \mu_t(S_t))^\top \beta_1) \right)^2 \right]. \quad (7)$$

The resulting minimizer is denoted as $\hat{\beta}_0^{\text{A2}}$, our A2-WCLS estimator. We can also consider including Z_t as a control variable in the $g_t(H_t)^\top \alpha$ model. However, since $g_t(H_t)^\top \alpha$ serves as a working model for $\mathbb{E}[W_t Y_{t+1} | H_t]$, it is expected to capture sufficient information. To avoid redundancy, we assume $Z_t \subseteq g_t(H_t)$.

An appropriate choice of the centering function $\mu_t(S_t)$ is key to obtaining a consistent estimation of the moderated causal effect. We must ensure that including the additional interaction term does not bias estimation of β_0 . That is, the minimizers of two objective functions, one in (7) and the other in (3), with and without auxiliary variable adjustment, should both converge asymptotically to the same target value β_0^* . This can be formulated as a set of linear constraints on the centering function $\mu_t(S_t)$:

Condition 3.4 (The orthogonality condition). For the i th component of $Z_t \in \mathbb{R}^p$, denoted $Z_t^i \in \mathbb{R}$, $i = 1, \dots, p$, the associated centering function, denoted $\mu_t^i(S_t)$, must satisfy the following orthogonality condition with respect to $f_t(S_t) \in \mathbb{R}^q$:

$$\mathbb{E} \left[\sum_{t=1}^T \tilde{p}_t(1 | S_t) (1 - \tilde{p}_t(1 | S_t)) (Z_t^i - \mu_t^i(S_t)) f_t(S_t) \right] = \mathbf{0}_{q \times 1}. \quad (8)$$

where $\tilde{p}_t(1 | S_t) = \tilde{p}_t(A_t = 1 | S_t)$ is the numerator of the weight W_t that is an arbitrary function with range $(0, 1)$ depending only on S_t .

Recognizing that β_1 is a nuisance parameter in (7) when estimating β_0 , we draw on the Neyman orthogonality of estimating equations to establish the above condition (Neyman 1979; Chernozhukov et al. 2018). Condition 3.4 leads to a low-dimensional centering function $\mu_t^i(S_t)$ that applies uniformly across all time points. This is our key technical contribution, showing how to incorporate time-varying auxiliary variables without compromising the statistical consistency of the estimator. We illustrate the utility of Condition 3.4 by considering an initial, simple example:

Example 3.5 (Fully marginal causal effect estimation). Consider estimating the fully marginal causal excursion effect β_0^* , where the primary moderator set is null (i.e., $S_t = \emptyset$ and $f_t(S_t) = 1$). Our goal is to incorporate auxiliary variable $Z_t \in \mathbb{R}$ to improve the estimation. Here we compare three centering function choices: (a) $\mu_t = \bar{Z}_t$, ($t = 1, 2, \dots, T$), (b) $\mu_t = \mu = \bar{Z}$, and (c) $\mu_t = \mu = \frac{\sum_{i=1}^T \hat{p}_t(1|S_t)(1-\hat{p}_t(1|S_t))Z_t}{\sum_{i=1}^T \hat{p}_t(1|S_t)(1-\hat{p}_t(1|S_t))}$. Option (a) estimates the time-specific mean $\mathbb{E}[Z_t]$ following the approach introduced by Lin (2013), indicating a correct model specification. It satisfies (8), but involves T estimates, which can be cumbersome when T is large. Option (b), the global mean of the auxiliary variable $\{Z_t\}_{t=1}^T$, is a scalar centering function but can lead to inconsistent estimates of β_0^* since it does not satisfy (8). Please refer to Appendix K.1 for an empirical illustration of the biased estimate. Option (c) is a scalar centering function satisfying (8), representing a weighted average of $\{Z_t\}_{t=1}^T$. In conclusion, option (c) has the desirable characteristics we seek—a low-dimensional centering function that is uniformly applicable across all time points and consistently estimates the causal parameter β_0^* .

This example shows that the consistency of $\hat{\beta}_0$ does not depend on the specific choice of $\mu_t^i(S_t)$, as long as it satisfies Condition 3.4. This flexibility allows $\mu_t^i(S_t)$ to extend beyond the standard conditional means $\mathbb{E}[Z_t^i|S_t]$, which are often difficult to estimate consistently, even in moderate dimensions, and increase in number with T . Equation (8), forming a system of q linear equations, offers a simpler alternative by suggesting that $\mu_t^i(S_t)$ can be defined as a time-smoothed linear projection of the auxiliary variable Z_t^i onto the subspace defined by $f_t(S_t) \in \mathbb{R}^q$. This reduces the nuisance parameters from T to the q coefficients of the linear model. Since q is generally small to maintain interpretability, this approach avoids the prohibitive growth of nuisance parameters with T while ensuring consistent estimation of the causal parameter of interest.

To illustrate the proposal more concretely, we can express the centering function as $\mu_t^i(S_t; \theta^i) = f_t(S_t)^\top \theta^i$, where $i \in \{1, \dots, p\}$, and $\hat{\theta}^i \in \mathbb{R}^q$ can be computed by solving (8). Let $\hat{\Theta} = (\hat{\theta}^1, \dots, \hat{\theta}^p) \in \mathbb{R}^{q \times p}$, we can construct an easy-to-implement A2-WCLS criterion as follows:

$$\mathbb{P}_N \left[\sum_{t=1}^T W_t \times \left(Y_{t+1} - g_t(H_t)^\top \alpha - (A_t - \tilde{p}_t(1|S_t)) (f_t(S_t)^\top \beta_0 + (Z_t - \hat{\Theta}^\top f_t(S_t))^\top \beta_1) \right)^2 \right]. \quad (9)$$

The following lemma describes the asymptotic properties of the A2-WCLS estimator, allowing us to compare its estimation precision with the existing WCLS estimator.

Lemma 3.6. Suppose Assumptions 2.1 and 3.1 hold, centering function $\{\mu_t(S_t)\}_{t=1}^T$ satisfies the Orthogonality Condition 3.4, and that the randomization probability $p_t(A_t = 1|H_t)$ is known. Let $(\hat{\alpha}, \hat{\beta}_0^{A2}, \hat{\beta}_1)$ minimize objective function (7). Under regularity conditions, $\hat{\beta}_0^{A2}$ is consistent and asymptotically normal such that $\sqrt{N}(\hat{\beta}_0^{A2} - \beta_0^*) \rightarrow \mathcal{N}(0, Q^{-1} \Sigma^{A2} Q^{-1})$, where Q, Σ^{A2} are defined in Appendix E.

Here we consider T as a fixed value, with the sample size N approaching infinity. In some cases, MRTs may involve small sample sizes compared to the scale of total time points T . Our simulations in Section 6 addressed this by applying a small sample correction (Mancl and DeRouen 2001) to the robust variance estimator, which yielded strong empirical performance for small sample sizes.

3.2.2. Post-Treatment Auxiliary Variable Adjustment

There is increasing interest in examining how current treatments affect lagged outcomes ($\Delta > 1$). In this context, incorporating auxiliary variables follows similar principles as with proximal outcomes but provides a broader set of options. In particular, *post-treatment auxiliary variables*, which are often highly predictive, can be included to help reduce noise and improve the precision of causal parameter estimates.

All variables included in (3), as indicated by the notation, must be observed prior to the treatment allocation at time t (A_t). Motivated by the efficient influence function of β_0 in Murphy et al. (2001), we propose to improve the estimator in (3) by subtracting its projection onto the score functions of the nuisance parameters (Bickel et al. 1993; Robins 1999), specifically the treatment randomization probabilities between $t+1$ and $t+\Delta-1$. This adjustment has the potential to enhance efficiency; however, it inherently incorporates post-treatment variables, which, if not carefully accounted for, may introduce bias, as demonstrated in Appendix F.1.

To address this challenge, we introduce the following A2-WCLS criterion for lagged outcomes, which effectively incorporates post-treatment auxiliary variables to enhance asymptotic efficiency, while mitigating bias from post-treatment adjustment. Let $l_{t+u}(A_{t+u}, H_{t+u})$ be a working model for $\mathbb{E}[W_{t+u, \Delta-1-u} Y_{t+\Delta} | A_{t+u}, H_{t+u}]$, $0 < u \leq \Delta-1$. Assuming $\pi_{t+u} = p_{t+u}$, it follows that the weights simplify to $W_{t+u} = 1$ for all u . We define the centering functions as $\mu_{t+u}^l(H_{t+u}) = \mathbb{E}[l_{t+u}(A_{t+u}, H_{t+u}) | H_{t+u}] = \sum_{a_{t+u}} p_{t+u}(a_{t+u} | H_{t+u}) l_{t+u}(a_{t+u}, H_{t+u})$. The A2-WCLS estimator for the Δ -lag causal effect, $\hat{\beta}_0^{A2}$, is obtained by minimizing:

$$\mathbb{P}_N \left[\sum_{t=1}^{T-\Delta+1} W_t \left(Y_{t+\Delta} - \sum_{u=1}^{\Delta-1} (l_{t+u}(A_{t+u}, H_{t+u}) - \mu_{t+u}^l(H_{t+u})) - \mu_t^l(H_t) - (A_t - \tilde{p}_t(1|S_t)) (f_t(S_t)^\top \beta_0 + (Z_t - \mu_t(S_t))^\top \beta_1) \right)^2 \right]. \quad (10)$$

The objective function (10) preserves the robust estimation property of WCLS, ensuring a consistent estimate of β_0 even if $l_{t+u}(A_{t+u}, H_{t+u})$ is misspecified (Boruvka et al. 2018). In the simple case where $\Delta = 2$, we can specify

$l_{t+u}(A_{t+u}, H_{t+u})$ as a linear working model $(A_{t+u} - p_{t+u})(\alpha_{u,1} + f_{t+u}(S_{t+u})^\top \alpha_{u,2}) + f_{t+u}(S_{t+u})^\top \alpha_{u,3}$. This leads to the centering function $\mu_{t+u}^l(H_{t+u}) = f_{t+u}(S_{t+u})^\top \alpha_{u,3}$. As a result, (10) can be expressed as

$$\begin{aligned} \mathbb{P}_N \left[\sum_{t=1}^{T-\Delta+1} W_t \left(Y_{t+\Delta} - ((A_{t+1} - p_{t+1}) \right. \right. \\ \left. \left. (\alpha_{1,1} + f_{t+1}(S_{t+1})^\top \alpha_{1,2}) \right) \right. \\ \left. - f_t(S_t)^\top \alpha_{0,3} - (A_t - \tilde{p}_t(1 | S_t))(f_t(S_t)^\top \beta_0 \right. \\ \left. + (Z_t - \mu_t(S_t))^\top \beta_1) \right)^2 \Big]. \end{aligned} \quad (11)$$

This linear working model simplifies the inference procedure, as the estimation of nuisance parameters $(\alpha_{u,1}, \alpha_{u,2}) \in \mathbb{R}^{q+1}$ does not introduce additional variance into the estimation of the causal parameter β_0 . The following lemma shows the asymptotic properties of the A2-WCLS estimator for the Δ -lag causal effect. Proofs can be found in Appendix F.

Lemma 3.7. Under the conditions of Lemma 3.6 and the reference treatment probability satisfies $\pi_{t+u} = p_{t+u}$ for all $0 < u \leq \Delta - 1$. Assuming $\beta_{p,\pi}(t + \Delta - 1; s) = f_t(s)^\top \beta_0^*$, and let $\hat{\beta}_0^{A2}$ minimize objective function (10). Under regularity conditions, $\hat{\beta}_0^{A2}$ is consistent and asymptotically normal such that $\sqrt{N}(\hat{\beta}_0^{A2} - \beta_0^*) \rightarrow \mathcal{N}(0, Q^{-1} \Sigma_l^{A2} Q^{-1})$, where Q, Σ_l^{A2} are defined in Appendix F.

In the more general case where $\pi_{t+u} \neq p_{t+u}$, the estimating equation becomes more complex, and we defer it to Appendix F.

4. Theoretical Results

4.1. Efficiency Comparison of the Proposed Estimator

We have demonstrated the sufficient condition for obtaining consistent estimates when incorporating auxiliary variables. This section explores the estimation efficiency for time-varying causal effect moderation. Since treatments, moderators, and outcomes exhibit temporal dependence, and the estimators for moderated effects are smoothed functions over time, a global efficiency gain, such as that shown in Lemma 3.3 for time-specific causal effects or in the theorem from Lin (2013), is no longer guaranteed. However, local efficiency gains can be achieved under additional assumptions specific to modeling time-varying moderated causal effects.

Assumption 4.1 (Correct causal model specification). The true causal effect model $\beta_{p,\pi}(t + \Delta - 1; H_t)$, $\Delta \geq 1$ is correctly specified in H_t as a linear model.

Here, we do not impose specific distribution assumptions on potential outcomes $\{Y_{t+\Delta}(\bar{A}_{t-1}, a), a \in \{0, 1\} \text{ and } \Delta \geq 1\}$, nor require the true data-generating model $\mathbb{E}[Y_{t+\Delta} | H_t, A_t]$ to be correctly specified. We only require the difference in conditional outcomes satisfy $\beta(t + \Delta - 1; H_t) = \mathbb{E}[W_{t,\Delta-1} Y_{t+\Delta} | H_t, 1] - \mathbb{E}[W_{t,\Delta-1} Y_{t+\Delta} | H_t, 0] = f_t(S_t)^\top \beta_0^* + (Z_t - \mu_t(S_t))^\top \beta_1^*$.

Assumption 4.2 (Exogenous residuals). The residuals in the outcomes and treatments, not accounted for by the observed his-

tory, is conditionally independent of the future observations given the complete history.

This assumption implies that the residuals do not carry any additional information beyond that which is already contained in the observed history. This is fundamental in many statistical learning models, particularly in Markov Decision Processes (MDPs), where the current state determines future states, independent of any unobserved information. Based on the assumptions outlined above, we can state the following theorem:

Theorem 4.3. For estimating the Δ -lag causal excursion effect, A2-WCLS offers the following efficiency properties compared to the WCLS estimator (Boruvka et al. 2018):

1. Under the conditions of Lemma 3.6 and Assumptions 4.1 and 4.2, incorporating auxiliary variables via (7) with a known centering function $\mu_t(S_t)$ guarantees an asymptotic efficiency improvement over the WCLS estimator in estimating the moderated causal effect for a given $S_t = s$, by $f_t(s)^\top Q^{-1} \Sigma Q^{-1} f_t(s)$, where Q, Σ are defined in Appendix G.
2. Under the conditions of Lemma 3.7 and Assumptions 4.1, if $\{l_{t+u}(H_{t+u}, A_{t+u})\}_{u=0}^{\Delta-1}$ are correctly specified, incorporating post-treatment auxiliary variables via (10) achieves semiparametric efficiency for estimating the Δ -lag moderated causal effect. The derivative of (10) with respect to β_0 is the efficient influence function for β_0 .

Although an efficiency gain is not generally guaranteed when the causal effect models are misspecified, the following proposition shows that under certain conditions, adjusting for auxiliary variables using the proposed A2-WCLS estimating equation ensures that the asymptotic variance of the causal parameter estimates does not increase:

Proposition 4.4. If the auxiliary variables Z_t is conditionally mean independent of the causal excursion effect given S_t , then incorporating them via A2-WCLS yields an estimator with the same asymptotic variance as the standard WCLS estimator (Boruvka et al. 2018).

Recall from (2) that $\beta(t; S_t, Z_t)$ is defined as $\mathbb{E}[\mathbb{E}[Y_{t+1} | H_t, 1] - \mathbb{E}[Y_{t+1} | H_t, 0] | S_t, Z_t]$. A straightforward derivation shows that this condition is equivalent to $\beta_1^* = 0$. To provide context, when selecting auxiliary variables for adjustment, there is a risk of including irrelevant ones. However, this proposition shows that even if non-effect moderators are mistakenly included, the asymptotic variance of the estimator will not be inflated using the A2-WCLS estimating equation.

4.2. Estimating the Centering Functions

Theorem 4.3 assumes known centering functions $\mu_t(S_t)$ and conditional mean functions $l_{t+u}(H_{t+u}, A_{t+u})$; however, in practice, these functions may need to be estimated from observed data, and analysis should account for this extra variance to ensure proper uncertainty quantification. A “stacked estimating equations” approach, a type of M-estimator (Carroll et al. 2006), is adopted to construct a robust asymptotic variance estimator. We parameterize the working models for $\mu_t(S_t)$

and $l_{t+u}(H_{t+u}, A_{t+u})$ with θ_μ and θ_l , respectively, and define $\Theta = (\theta_\mu, \theta_l)$. The corresponding estimating equation is denoted as $U(\Theta)$. The estimating equation for β_0 , denoted as $U_{\beta_0}(\beta_0; \hat{\Theta}, \hat{\beta}_1)$, is the derivative of the proposed A2-WCLS criterion in (7) with respect to β_0 :

$$U_{\beta_0}(\beta_0; \hat{\Theta}, \hat{\beta}_1) = \mathbb{P}_N \left[\sum_{t=1}^T W_t \left(Y_{t+\Delta} - \sum_{u=1}^{\Delta-1} (l_{t+u}(A_{t+u}, H_{t+u}; \hat{\theta}_l) - \mu_{t+u}^l(H_{t+u}; \hat{\theta}_l)) - \mu_t^l(H_t; \hat{\theta}_l) - (A_t - \tilde{p}_t(1|S_t))(f_t(S_t)^\top \beta_0 + (Z_t - \mu_t(S_t, \hat{\theta}_\mu)^\top \hat{\beta}_1)(A_t - \tilde{p}_t)f_t(S_t)) \right) \right], \quad (12)$$

where $\hat{\beta}_1$ is obtained by minimizing (7), and $\hat{\Theta}$ is obtained from solving $U(\Theta) = 0$. By applying the stacked estimating equations approach, we establish the following asymptotic property for the causal parameter estimate $\hat{\beta}_0^{A2}$:

Lemma 4.5. Under the conditions of Lemmas 3.6 or 3.7, if we incorporate auxiliary variables via A2-WCLS with estimated functions $\mu_t(S_t, \hat{\theta}_\mu)$ and $l_{t+u}(A_{t+u}, H_{t+u}; \hat{\theta}_l)$, where $\hat{\Theta}$ is obtained from solving $U(\Theta) = 0$, then $\hat{\beta}_0^{A2}$ is consistent and asymptotically normal such that $\sqrt{N}(\hat{\beta}_0^{A2} - \beta_0^*) \rightarrow \mathcal{N}(0, Q^{-1} \tilde{\Sigma}^{A2} Q^{-1})$, where Q is as defined in Theorem 4.3, and $\tilde{\Sigma}^{A2} = \mathbb{E}[\tilde{U}_{\beta_0}(\hat{\beta}_0; \hat{\Theta}, \hat{\beta}_1)^\top \tilde{U}_{\beta_0}(\hat{\beta}_0; \hat{\Theta}, \hat{\beta}_1)]$, in which $\tilde{U}_{\beta_0}(\hat{\beta}_0; \hat{\Theta}, \hat{\beta}_1)$ is obtained as follows:

$$\tilde{U}_{\beta_0}(\hat{\beta}_0; \hat{\Theta}, \hat{\beta}_1) = U_{\beta_0}(\hat{\beta}_0; \hat{\Theta}, \hat{\beta}_1) - U(\hat{\Theta}) \left(\mathbb{E} \left[\frac{\partial U(\Theta)}{\partial \Theta} \Big|_{\Theta=\hat{\Theta}} \right] \right)^{-1} \mathbb{E} \left[\frac{\partial U_{\beta_0}(\hat{\beta}_0; \Theta, \hat{\beta}_1)}{\partial \Theta} \Big|_{\Theta=\hat{\Theta}} \right]^\top. \quad (13)$$

While the efficiency improvement may be smaller compared to Theorem 4.3, our proposed method remains robust to the estimated centering functions. We see empirical finite-sample efficiency gains in Section 6, and provide a further discussion in Appendix I.

5. Properties

In this section, we return to the five desired properties in the checklist discussed in Section 2.2 and demonstrate that A2-WCLS satisfies all of them.

Valid statistical inference. As demonstrated in Lemma 3.6, 3.7, and Theorem 4.3, our approach ensures consistent and asymptotically normal estimation of the causal parameter. Moreover, under Assumptions 4.1 and 4.2, A2-WCLS has theoretical guarantees for achieving an efficiency gain relative to the original WCLS method.

Robust estimation. (1) Proposition 4.4 indicates that the inclusion of non-effect-moderator auxiliary variables in the causal effect model does not compromise the precision of the estimation of the parameter of interest. (2) The A2-WCLS method shares the same property as WCLS in that if the true random-

ization probability $p_t(A_t|H_t)$ is known, a consistent estimator can be obtained even when the nuisance function $g_t(H_t)^\top \alpha$ and $l_{t+u}(H_{t+u}, A_{t+u})$ are misspecified (Boruvka et al. 2018). Robust standard errors are used to account for model misspecification and heteroscedasticity.

Strong finite-sample performance. Our conclusions are derived based on asymptotic properties of the estimators. However, our proposed method can still provide reliable performance in a finite sample with a small sample correction (Mancl and DeRouen 2001) applied to the variance estimator, which is elaborated in Appendix J.

Wide applicability. The proposed method is a rigorous longitudinal extension of the existing literature on covariate adjustment that can be applied to study designs with $T \geq 1$, as discussed in Lemma 3.3 and Theorem 4.3. When $T = 1$, the adjustment of the auxiliary variables is based on baseline information, subsuming the covariate adjustment techniques discussed in Lin (2013) and Ye et al. (2023) as special cases.

Computational simplicity. Using linear working models $\Theta^\top f_t(S_t)$ for the centering function simplifies computation, and can be easily implemented using existing routines found in standard statistical software packages. See the reproducible code and documentations provided with this article.

6. Simulation

Here, the empirical performance of the proposed A2-WCLS method is examined through simulation. Results verify the theory that the A2-WCLS method can significantly improve the efficiency of causal effect estimation, even when accounting for the additional variance introduced by estimating the centering function from data.

6.1. Simulation Setup

The proposed method is empirically evaluated via simulation, where the data generation model is based on Boruvka et al. (2018) with minor adjustments made to demonstrate the improvement of our proposed method over the WCLS approach. Consider an MRT with known randomization probability and observation vector being a single state variable $Z_t \in \{-1, 1\}$ with probability 0.5 taking each value at each decision time t , Let

$$\begin{aligned} Y_{t+2,j} = & 0.2Z_{t+1,j} + (-0.2 + 0.8(Z_{t+1,j} - \mathbb{E}[Z_{t+1,j}])) \\ & \times (A_{t+1,j} - p_{t+1}(1|H_{t+1,j})) + \\ & (\beta_0 + \beta_1(Z_{t,j} - \mathbb{E}[Z_{t,j}])) \\ & \times (A_{t,j} - p_t(1|H_{t,j})) + \epsilon_{t,j}. \end{aligned} \quad (14)$$

The randomization probability is set to $p_t(1|H_t) = \text{expit}(\eta_1 A_{t-1} + \eta_2 Z_t)$ where $(\eta_1, \eta_2) = (-0.8, 0.8)$ and $\text{expit}(x) = (1 + \exp(-x))^{-1}$. The independent error term satisfies $\epsilon_t \sim \mathcal{N}(0, 1)$ with $\text{corr}(\epsilon_u, \epsilon_t) = 0.5^{|u-t|/2}$. We set $\beta_0 = -0.1$, and $\beta_1 \in \{0.2, 0.5, 0.8\}$ to represent a small, medium, and large lagged moderation effect.

6.2. Simulation Results

In the main simulation experiment, the lag $\Delta = 2$ marginal causal effect is estimated, which is constant in time and is given by $\beta_0^* = -0.1$. Results are reported with $N = 250$ individuals and $T = 30$ using the following estimation methods:

Estimation Method I: WCLS. The WCLS method (Boruvka et al. 2018) is applied using a fully marginal model, $Y_{t+2} \sim A_t$. This approach ensures a consistent estimate with a valid confidence interval. Therefore, we use it as a benchmark for comparing the estimation results of the proposed method.

Estimation Method II: A2-WCLS. In our simulation study, we applied the A2-WCLS method as specified in (12). The pre-treatment auxiliary variable used for adjustment was $Z_{t,j}$. Since our primary interest was estimating the fully marginal causal excursion effect (i.e., $S_t = \emptyset$, $f_t(S_t) = 1$), we set the working model for the centering function under the A2-WCLS criterion in (9) as $\mu_t(S_t) = \theta$, with the corresponding estimate given by $\hat{\theta} = \frac{\mathbb{P}_n[\sum_{t=1}^T \tilde{p}_{t,j}(1-\tilde{p}_{t,j})Z_{t,j}]}{\mathbb{P}_n[\sum_{t=1}^T \tilde{p}_{t,j}(1-\tilde{p}_{t,j})]}$. In addition, for post-treatment auxiliary variable adjustment, we used the working model $l_{t+1}(A_{t+1,j}, H_{t+1,j}) = (A_{t+1,j} - p_{t+1,j})(\alpha_{1,1} + Z_{t+1,j}\alpha_{1,2}) + Z_{t+1,j}\alpha_{1,3}$.

The simulation experiment involves estimating θ using the data generated, and this introduces non-negligible variation that affects the asymptotic variance estimation for the fully marginal causal effect. Therefore, the asymptotic variance estimator shown in (13) is used. Table 2 reports the simulation results. “%RE gain” indicates the percentage of times we achieve an efficiency gain out of 1000 Monte Carlo replicates. “mRE” represents the average asymptotic relative efficiency (i.e., an asymptotic variance ratio), and “RSD” represents the relative standard deviation between two estimates.

Despite having to account for the extra variance caused by estimating θ , the proposed A2-WCLS method still significantly improves the efficiency of fully marginal lagged causal effect estimation. Moreover, as β_1 increases, indicating greater causal model misspecification in the WCLS estimator, A2-WCLS achieves even larger efficiency gains due to its ability to incorporate pre-treatment auxiliary variables that are strong effect moderators, effectively reducing estimation variability.

Except for estimating the fully marginal lagged effect, we conducted additional simulations to assess the performance of the proposed A2-WCLS estimator. A simulation evaluating the causal excursion effect for the proximal outcome based on (7) is provided in Appendix K.2. Further evaluations of A2-WCLS in estimating time-varying treatment effects are outlined in Appendix K.3. The results demonstrate consistent estimation

and efficiency gains. In Appendix K.4, we explore how efficiency gains vary with different choices for sample size N and total time points T .

7. Application to Intern Health Study

The Intern Health Study (IHS) is a six-month microrandomized trial on medical interns (NeCamp et al. 2020), which aimed to investigate when to provide mHealth interventions to people in stressful work environments to improve their behavior and mental health. In this section, we evaluate the effectiveness of targeted notifications in improving an individual’s mood and step counts. The dataset contains 1562 participants.

The exploratory and MRT analyses conducted in this article focus on weekly randomization, thus, an individual was randomized to receive mood, activity, sleep, or no notifications with equal probability (1/4 each) every week. We choose the outcome $Y_{t+1,j}$ as the average self-reported mood score (a Likert scale taking values from 1 to 10) and step count (cubic root transformed) for individual j in study week t . The average weekly mood score when a notification is delivered is 7.14, and 7.16 when there is no notification; The average weekly step count (cubic root) when a notification is delivered is 19.1, and also 19.1 when there is no notification. In the following analysis, the prior week’s outcome is chosen as the auxiliary variable. We evaluate the targeted notification treatment effect for medical interns using our proposed method A2-WCLS and WCLS.

7.1. Comparison of the Marginal Effect Estimation

First, the fully marginal excursion effect is assessed (i.e., $\beta(t) = \beta_0^*$). For an individual j , the study week is coded as a subscript t . $Y_{t+1,j}$ is the self-reported mood score or step count of the individual j in study week $t + 1$. $A_{t,j}$ is defined as the specific type of notification that targets improving the outcome. For example, if the outcome is the self-reported mood score, sending mood notifications would be the treatment, thus, $\mathbb{P}(A_{t,j} = 1) = 0.25$. We analyze the fully marginal causal effect of the targeted notifications on self-reported mood score and step count.

We considered three estimators for the fully marginal treatment effect β_0^* . Estimator I is the unadjusted estimator that uses only the time-varying outcome and treatment assignment data. Estimator II is obtained using the WCLS method (Boruvka et al. 2018) with the outcome of the previous week $Z_{t,j} := Y_{t,j}$ as a control variable. Specifically, for mood score outcomes, we selected the previous week’s average mood score as the control variable, and for step count outcomes, we chose the prior week’s average step count. Estimator III is obtained by further adjusting $Z_{t,j}$ as an auxiliary variable using the proposed A2-WCLS method, with the working centering function defined as $\mu_t = \mathbb{P}_N[Z_{t,j}] = \mathbb{P}_N[Y_{t,j}]$.

We report the estimates in Figure 1 and present more details in Table S7 of Appendix M. Compared to Estimator I, Estimators II & III have a tangible improvement in standard error estimates, with a relative estimation efficiency of 2.32, 2.33 for the mood outcome and 1.57, 1.58 for the step outcome. Even though Estimator III using A2-WCLS does not gain considerable efficiency over Estimator II (WCLS) results, it is not likely to lose efficiency.

Table 2. Fully marginal lag $\Delta = 2$ causal effect estimation efficiency comparison.

Method	β_1	Est	SE	CP	%RE gain	mRE	RSD
WCLS	0.2	−0.100	0.030	0.946	–	–	–
	0.5	−0.100	0.032	0.943	–	–	–
	0.8	−0.100	0.033	0.941	–	–	–
A2-WCLS	0.2	−0.100	0.028	0.939	99.6%	1.141	1.054
	0.5	−0.100	0.029	0.937	99.8%	1.161	1.064
	0.8	−0.100	0.031	0.943	100%	1.168	1.070

NOTE: The true value of the parameters is $\beta_0^* = -0.1$.

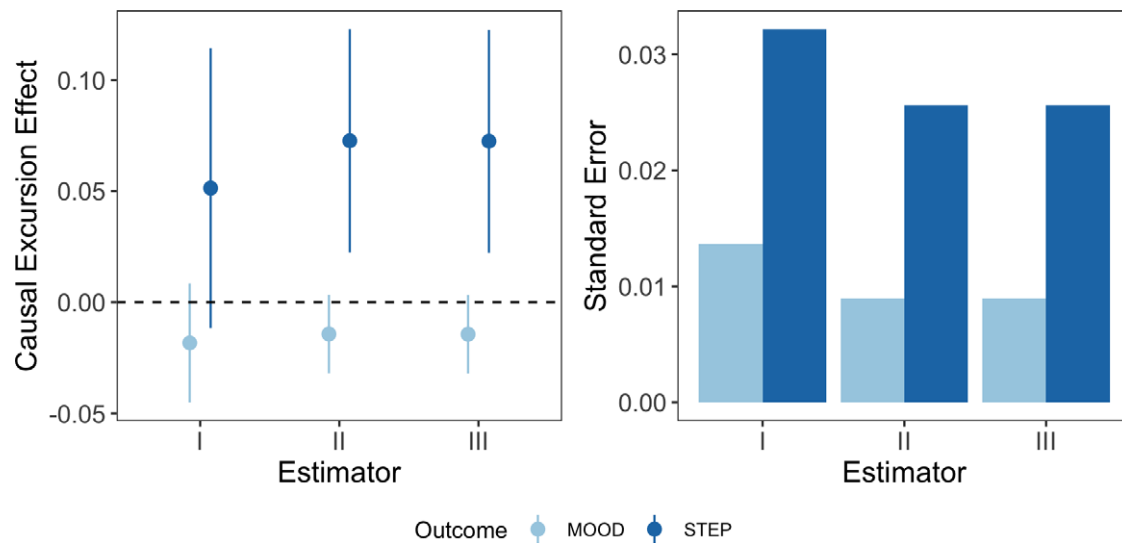


Figure 1. Causal effect estimates with the 95% confidence interval, and standard errors.

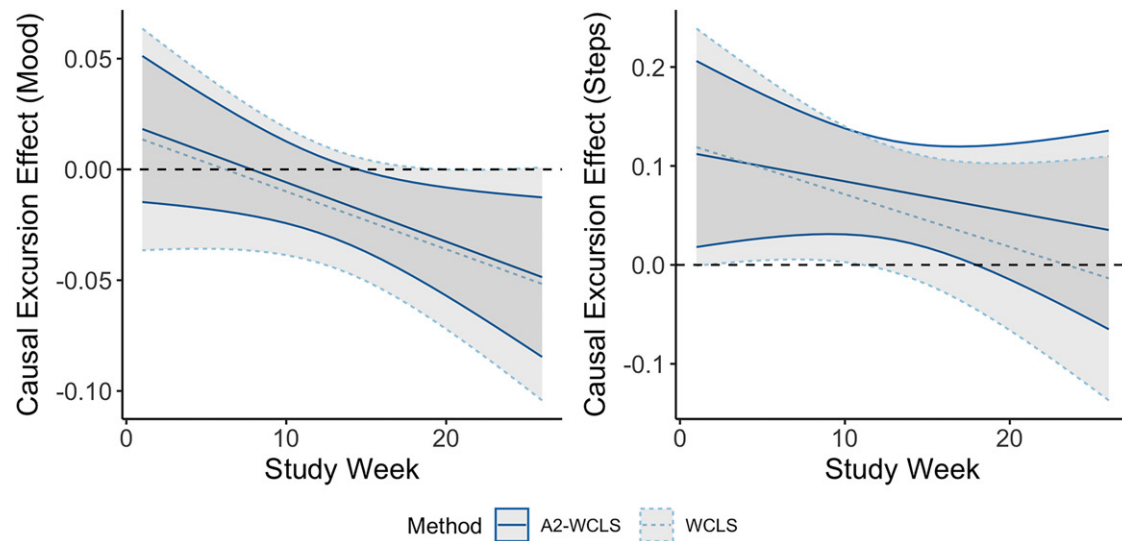


Figure 2. Causal effect estimates with the 95% confidence interval.

We conclude that sending activity notifications can increase (the cubic root of) step counts by 0.072, with statistical significance at level 95%. However, we do not observe a significant effect of sending mood notifications on users' moods.

7.2. Time-Varying Treatment Effect Estimation

In most mobile health intervention studies, time-in-study has been an important moderator of treatment effects due to habituation or learning effect. Therefore, we include study week in the marginal treatment effect model for further investigation: $\beta(t) = \beta_0^* + \beta_1^*t$. The auxiliary variables are still chosen to be the corresponding outcome in the previous week. Control variables have been established in Section 7.1 as being beneficial in reducing standard errors, so control variables (prior week's outcome) will always be included in the following analysis.

Figure 2 shows estimated time-varying treatment moderation effects. We compare our proposed approach against the WCLS method from Boruvka et al. (2018). More details of the

moderated analysis are presented in Table S8 of Appendix M. The shaded area represents the 95% confidence band of the moderation effects at varying values of the moderator (week in study). A narrower confidence band is observed when A2-WCLS is used; specifically, the relative efficiency for $\hat{\beta}_0, \hat{\beta}_1$ in the mood model is 2.204 and 2.157, respectively, and in the step model is 1.562 and 1.475, respectively.

There is an overall decreasing trend in the effectiveness of the treatments on both proximal outcomes with study week. The scientific rationale for this trend may be habituation to or burden caused by the targeted reminders. Furthermore, the causal excursion effect of mobile prompts for step count change is positive and statistically significant in the first several weeks of the study. This suggests that sending targeted reminders may be beneficial to increase physical activity in the short term. Note that the estimated effect using A2-WCLS is significant until week 18, while using WCLS the effect is only significant until week 12. This demonstrates the importance of including auxiliary variables for improved precision in causal moderation analysis.

Both analyses show that in later stages of the study, the effect is no longer significant, which may be due to habituation to and/or burden caused by the smartphone reminders.

8. Discussion

MRTs are frequently adopted in mobile health studies to assess the effectiveness of just-in-time intervention components. Although abundant auxiliary variables are often collected in these studies through wearables and smartphones, their statistical utility for enhancing causal effect analysis has not kept pace with its counterpart in classical RCTs. To address this gap, we propose the A2-WCLS approach, which integrates auxiliary variables into the estimation of moderated causal excursion effects. Our work tackles the long-standing questions of (a) how to leverage auxiliary information, (b) when it provides benefits, and (c) the extent of those benefits in the context of repeated measurements and time-varying treatment effects, subsuming existing popular covariate adjustment methods in classical RCTs as a special case (Section 3).

Under some interpretable conditions in time-varying settings, the proposed A2-WCLS method offers improved efficiency for moderated causal excursion effect estimation in comparison to the benchmark WCLS methods. In Appendix N, we provide clear guidelines for practitioners to implement auxiliary variable adjustment in longitudinal studies.

Many open questions remain. First, we may use feature selection methods to regularize the centering function $\mu_t(S_t)$, for example, by the smooth clipped absolute deviation method (SCAD) (Fan and Li 2001), to pick an optimal model for $\mu_t(S_t)$. A nonlinear centering function can also be considered, similar to Guo and Basse (2023) proposed for RCTs. In addition to using auxiliary variables to improve estimation efficiency. Third, if the MRT has a clustered structure (interference within clusters and/or treatment heterogeneity at the cluster level), it requires new centering techniques using cluster-level auxiliary variables (Shi, Wu, and Dempsey 2023). Finally, more advanced and flexible MRT designs should be explored so that they may use auxiliary variable adjustment to maximize efficiency. For example, Van Lancker, Betz, and Rosenblum (2022) proposes using an information adaptive design, which adapts to the amount of precision gain and can lead to faster, more efficient trials, without sacrificing validity or power. We leave these directions for future work.

Disclosure Statement

The authors report there are no competing interests to declare.

References

- Bickel, P. J., Klaassen, C. A., Bickel, P. J., Ritov, Y., Klaassen, J., Wellner, J. A., and Ritov, Y. (1993), *Efficient and Adaptive Estimation for Semiparametric Models* (Vol. 4), New York: Springer. [305]
- Boruvka, A., Almirall, D., Witkiewitz, K., and Murphy, S. A. (2018), "Assessing Time-Varying Causal Effect Moderation in Mobile Health," *Journal of the American Statistical Association*, 113, 1112–1121. [300,301,302,303,304,305,306,307,308,309]
- Carroll, R. J., Ruppert, D., Stefanski, L. A., and Crainiceanu, C. M. (2006), *Measurement Error in Nonlinear Models: A Modern Perspective, Second Edition*, New York: Chapman and Hall/CRC. [306]
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018), "Double/Debiased Machine Learning for Treatment and Structural Parameters," *The Econometrics Journal*, 21, C1–C68. [305]
- Dempsey, W., Liao, P., Klasnja, P., Nahum-Shani, I., and Murphy, S. A. (2015), "Randomised Trials for the Fitbit Generation," *Significance*, 12, 20–23. [300]
- Dempsey, W., Liao, P., Kumar, S., and Murphy, S. A. (2020), "The Stratified Micro-Randomized Trial Design: Sample Size Considerations for Testing Nested Causal Effects of Time-Varying Treatments," *The Annals of Applied Statistics*, 14, 661–684. [301,303]
- Eicker, F. (1967), "Limit Theorems for Regressions with Unequal and Dependent Errors," in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1), pp. 59–82, Berkeley, CA: University of California Press. [302]
- Fan, J., and Li, R. (2001), "Variable Selection via Nonconcave Penalized Likelihood and its Oracle Properties," *Journal of the American Statistical Association*, 96, 1348–1360. [310]
- Fisher, R. A. (1936), "Design of Experiments," *British Medical Journal*, 1, 554. [302]
- Free, C., Phillips, G., Galli, L., Watson, L., Felix, L., Edwards, P., Patel, V., and Haines, A. (2013), "The Effectiveness of Mobile-Health Technology-based Health Behaviour Change or Disease Management Interventions for Health Care Consumers: A Systematic Review," *PLoS Medicine*, 10, e1001362. [300]
- Freedman, D. A. (2006), "On the So-called 'Huber Sandwich Estimator' and 'Robust Standard Errors,'" *The American Statistician*, 60, 299–302. [302]
- (2008), "On Regression Adjustments to Experimental Data," *Advances in Applied Mathematics*, 40, 180–193. [302,303]
- Guo, K., and Basse, G. (2023), "The Generalized Oaxaca-Blinder Estimator," *Journal of the American Statistical Association*, 118, 524–536. [302,310]
- Heron, K. E., and Smyth, J. M. (2010), "Ecological Momentary Interventions: Incorporating Mobile Technology into Psychosocial and Health Behaviour Treatments," *British Journal of Health Psychology*, 15, 1–39. [301]
- Huber, P. J. (1967), "Under Nonstandard Conditions," in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability: Weather Modification*, University of California Press: Berkeley, CA, USA, pp. 221. [302]
- Kahan, B. C., Jairath, V., Doré, C. J., and Morris, T. P. (2014), "The Risks and Rewards of Covariate Adjustment in Randomized Trials: An Assessment of 12 Outcomes from 8 Studies," *Trials*, 15, 1–7. [300]
- Klasnja, P., Hekler, E. B., Shiffman, S., Boruvka, A., Almirall, D., Tewari, A., and Murphy, S. A. (2015), "Microrandomized Trials: An Experimental Design for Developing Just-in-Time Adaptive Interventions," *Health Psychology*, 34, 1220–1228. [300]
- Liang, K.-Y., and Zeger, S. L. (1986), "Longitudinal Data Analysis Using Generalized Linear Models," *Biometrika*, 73, 13–22. [302]
- Liao, P., Klasnja, P., Tewari, A., and Murphy, S. A. (2016), "Sample Size Calculations for Micro-Randomized Trials in mhealth," *Statistics in Medicine*, 35, 1944–1971. [301]
- Lin, W. (2013), "Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique," *The Annals of Applied Statistics*, 7, 295–318. [300,301,302,303,304,305,306,307]
- Mancl, L. A., and DeRouen, T. A. (2001), "A Covariance Estimator for GEE with Improved Small-Sample Properties," *Biometrics*, 57, 126–134. [305,307]
- Murphy, S. A., van der Laan, M. J., Robins, J. M., and C. P. P. R. Group. (2001), "Marginal Mean Models for Dynamic Regimes," *Journal of the American Statistical Association*, 96, 1410–1423. [305]
- Nahum-Shani, I., Smith, S. N., Spring, B. J., Collins, L. M., Witkiewitz, K., Tewari, A., and Murphy, S. A. (2018), "Just-in-Time Adaptive Interventions (jitais) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support," *Annals of Behavioral Medicine*, 52, 446–462. [300]
- NeCamp, T., Sen, S., Frank, E., Walton, M. A., Ionides, E. L., Fang, Y., Tewari, A., and Wu, Z. (2020), "Assessing Real-Time Moderation for Developing Adaptive Mobile Health Interventions for Medical Interns: Micro-Randomized Trial," *Journal of Medical Internet Research*, 22, e15033. [301,308]

- Negi, A., and Wooldridge, J. M. (2021), "Revisiting Regression Adjustment in Experiments with Heterogeneous Treatment Effects," *Econometric Reviews*, 40, 504–534. [300,302]
- Neyman, J. (1979), "C (α) Tests and their Use," *Sankhyā: The Indian Journal of Statistics*, Series A, 41, 1–21. [305]
- Neyman, J. S. (1923), "On the Application of Probability Theory to Agricultural Experiments. Essay on Principles. Section 9. (Translated and Edited by DM Dabrowska and TP Speed, *Statistical Science* (1990), 5, 465–480)," *Annals of Agricultural Sciences*, 10, 1–51. [302]
- Qian, T., Yoo, H., Klasnja, P., Almirall, D., and Murphy, S. A. (2021), "Estimating Time-Varying Causal Excursion Effects in Mobile Health with Binary Outcomes," *Biometrika*, 108, 507–527. [301,302]
- Robins, J. (1986), "A New Approach to Causal Inference in Mortality Studies with a Sustained Exposure Period—Application to Control of the Healthy Worker Survivor Effect," *Mathematical Modelling*, 7, 1393–1512. [301]
- Robins, J. M. (1994), "Correcting for Non-Compliance in Randomized Trials Using Structural Nested Mean Models," *Communications in Statistics-Theory and Methods*, 23, 2379–2412. [302]
- (1997), "Causal Inference from Complex Longitudinal Data," in *Latent Variable Modeling and Applications to Causality*, ed. M. Berkane, pp. 69–117, New York, NY: Springer. [302]
- (1999), "Testing and Estimation of Direct Effects by Reparameterizing Directed Acyclic Graphs with Structural Nested Models," in *Computation, Causation, and Discovery*, eds. G. F. Cooper and C. Glymour, pp. 349–405, Washington, DC: AAAI Press. [305]
- Rubin, D. (1978), "Bayesian Inference for Causal Effects: The Role of Randomization," *The Annals of Statistics*, 6, 34–58. [301]
- Shi, J., Wu, Z., and Dempsey, W. (2023), "Assessing Time-Varying Causal Effect Moderation in the Presence of Cluster-Level Treatment Effect Heterogeneity and Interference," *Biometrika*, 110, 645–662. [302,303,310]
- Su, F., and Ding, P. (2021), "Model-Assisted Analyses of Cluster-Randomized Experiments," *Journal of the Royal Statistical Society*, Series B, 83, 994–1015. [302]
- Tackney, M. S., Morris, T., White, I., Leyrat, C., Diaz-Ordaz, K., and Williamson, E. (2023), "A Comparison of Covariate Adjustment Approaches under Model Misspecification in Individually Randomized Trials," *Trials*, 24, 14. [300]
- Tsiatis, A. A., Davidian, M., Zhang, M., and Lu, X. (2008), "Covariate Adjustment for Two-Sample Treatment Comparisons in Randomized Clinical Trials: A Principled Yet Flexible Approach," *Statistics in Medicine*, 27, 4658–4677. [302]
- Van Lancker, K., Betz, J., and Rosenblum, M. (2022), "Combining Covariate Adjustment with Group Sequential and Information Adaptive Designs to Improve Randomized Trial Efficiency," arXiv preprint arXiv:2201.12921. [310]
- Vansteelandt, S., and Didelez, V. (2018), "Improving the Robustness and Efficiency of Covariate-Adjusted Linear Instrumental Variable Estimators," *Scandinavian Journal of Statistics*, 45, 941–961. [302]
- White, H. (2014), *Asymptotic Theory for Econometricians*, London: Academic Press Inc. Ltd. [302]
- Yang, L., and Tsiatis, A. A. (2001), "Efficiency Study of Estimators for a Treatment Effect in a Pretest–Posttest Trial," *The American Statistician*, 55, 314–321. [302]
- Ye, T., Shao, J., Yi, Y., and Zhao, Q. (2023), "Toward Better Practice of Covariate Adjustment in Analyzing Randomized Clinical Trials," *Journal of the American Statistical Association*, 118, 2370–2382. [300,302,307]